Column1	surgeon	mrn	patient_initials
12	Barrio, Andrea	39105473	NP
31	Capko, Deborah	38138814	BF
62	El-Tamer, Mahmoud	39106459	MW
63	El-Tamer, Mahmoud	39103258	MM
66	El-Tamer, Mahmoud	39111396	KS
70	El-Tamer, Mahmoud	39100321	DM
94	Heerdt, Alexandra	39095007	UD
103	Kirstein, Laurie	39100822	SW
117	Lee, Min	39094834	WY
134	Montag, Giacomo	39092180	DR
137	Montag, Giacomo	39089057	XM
144	Moo, Tracy-Ann	38307961	FS
163	Pawloski, Kate	39089385	MM
175	Plitas, George	39102467	MC
198	Tadros, Audree	39091629	RW

tumor_invasive_dcis	complex_case_status	lesion_size_status_source
2	0	1
1	1	1
2	0	1
2	0	1
1	0	1
1	1	1
1	0	1
1	0	1
1	0	1
1	0	1
1	0	1
2	0	1
1	0	1
1	0	1
1	0	1

lesion_size_status_human	lesion_size_status_ai	laterality_status_source	laterality_status_human
1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1
1	3	1	1
1	3	1	1
1	3	1	1
1	3	1	1
1	1	1	1
1	1	1	1
1	1	1	1
1	3	1	1
1	1	1	1
1	1	1	1
1	1	1	1

laterality_status_ai	lesion_location_status_source	lesion_location_status_human	lesion_location_status_ai
1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1

calcifications_asymmetry_status_source	calcifications_asymmetry_status_human	calcifications_asymmetry_status_ai	additional_enhancement_mri_status_source
1	1	1	1
0	NA	NA	0
1	1	1	1
1	1	1	0
0	NA	NA	0
1	1	1	1
1	1	1	0
1	1	1	1
1	1	1	0
0	NA	NA	0
1	1	1	0
0	NA	NA	0
0	NA	NA	0
0	NA	NA	0
1	1	1	0

additional_enhancement_mri_status_human	additional_enhancement_mri_status_ai	extent_status_source	extent_status_human
1	1	0	NA
NA	NA	1	2
1	1	1	1
NA	NA	1	1
NA	NA	0	NA
1	2	1	1
na	na	0	na
1	1	1	1
na	na	0	na
NA	NA	0	NA
NA	NA	1	1
NA	NA	0	NA
NA	NA	0	NA
NA	NA	0	NA
NA	NA	0	NA

extent_status_ai	accurate_clip_placement_status_source	accurate_clip_placement_status_human	accurate_clip_placement_status_ai	workup_recommendation_status_source	workup_recommendation_status_human
NA	1	1	3	1	1
1	1	2	1	0	NA
1	1	1	1	1	1
3	1	1	1	1	1
NA	1	1	1	1	2
1	1	1	1	1	1
na	1	1	1	1	1
1	1	1	1	1	1
na	1	1	1	1	1
NA	1	1	1	0	NA
2	0	NA	NA	0	NA
NA	1	1	1	0	NA
NA	1	1	1	1	1
NA	1	1	1	1	1
NA	1	1	1	1	1

workup_recommendation_status_ai	Lymph node_status_source	Lymph node_status_human	Lymph node_status_ai	chronology_preserved_status_source
1	1	1	1	1
NA	0	NA	NA	1
3	1	1	1	1
1	1	1	1	1
1	1	1	1	1
3	1	1	1	1
1	0	na	na	1
1	1	1	2	1
1	1	2	1	1
NA	1	1	1	1
NA	1	1	1	1
NA	1	1	1	1
1	1	1	1	1
1	0	NA	NA	1
1	1	1	1	1

chronology_preserved_status_human	chronology_preserved_status_ai	biopsy_method_status_source	biopsy_method_status_human	biopsy_method_status_ai	invasive_component_size_pathology_status_source
1	1	1	1	1	0
1	1	1	1	1	1
1	1	1	1	1	0
1	1	1	1	1	0
1	1	1	1	1	1
1	1	1	1	1	0
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	0
1	3	1	1	1	1
1	1	1	1	3	1
1	1	1	1	1	1

invasive_component_size_pathology_status_human	invasive_component_size_pathology_status_ai	histologic_diagnosis_status_source	histologic_diagnosis_status_human	histologic_diagnosis_status_ai	receptor_status_source
NA	NA	1	1	1	1
1	1	1	1	3	1
NA	NA	1	1	1	1
NA	NA	1	1	1	0
2	1	1	1	1	1
NA	NA	1	1	1	1
2	1	1	1	1	1
1	1	1	1	1	1
2	3	1	1	1	1
2	2	1	1	1	1
2	3	1	1	1	1
NA	NA	1	1	1	1
2	1	1	1	1	1
2	1	1	1	1	1
1	1	1	1	1	1

receptor_status_human	receptor_status_ai	comments	has_3_in_ai
1	1	clip not deployed/seen on post-biopsy images how	1
2	1	Ai read it as lobular reports say mammary	1
1	1	AI correctly wrote no Her2 in the receptors	1
NA	NA	in describing extent of disease from diag m	1
1	1	Size of tumor os US was 0.4, AI wrote 1.4,	1
1	1	MRI was available but AI did not mention i	1
1	2	missed receptors and reported incorrect si	1
1	1	Bot gave incorrect size for US mass 1.5 ins	1
1	1	Bot reports invasive component as 3.5mn	1
1	3	bot reports HR as TNBC, but triple positive	1
1	1	Poor quality PDF in which bot missed exte	1
1	1	Bot incorrectly reads initial left breast ultrasour	1
1	1		1
1	1	both human and AI HPI fail to mention no	1
1	3	bot got the receptors, stating ER- PR - whe	1

Validation Coding Legend

Source presence (*_status_source)

0 = Absent from source document

1 = Present in source document

Summary evaluation (*_status_human, *_status_ai)

NA = Not applicable (use when source = 0)

1 = Accurate representation (use clinical judgement)

2 = Omission (present in source but missing from summary)

3 = Fabrication (introduced or materially incorrect information)

Rule: If *_status_source = 0, both *_status_human and *_status_ai must be **NA**.

Lesion Size - as reresented across MMG/US/MRI, use clinincal judgement to determine if accurated represented
All columns refer to the index tumor/abnormality, leave note in comments for outliers
MRI enhancement refers to suspicious enhancement around index tumor or elsewhere in the breast
Clip placement accurate if clip type and displacement reported for all biopsied lesions
Biopsy method accurate if correct for all biopsied lesions

l by AI/Human in summary

Column1	surgeon	mrn	patient_initials	tumor_invasive_dcis
12	Barrio, Andrea	39105473	NP	2
31	Capko, Deborah	38138814	BF	1
62	El-Tamer, Mahmoud	39106459	MW	2
63	El-Tamer, Mahmoud	39103258	MM	2
66	El-Tamer, Mahmoud	39111396	KS	1
70	El-Tamer, Mahmoud	39100321	DM	1
94	Heerdt, Alexandra	39095007	UD	1
103	Kirstein, Laurie	39100822	SW	1
117	Lee, Min	39094834	WY	1
134	Montag, Giacomo	39092180	DR	1
137	Montag, Giacomo	39089057	XM	1
144	Moo, Tracy-Ann	38307961	FS	2
163	Pawloski, Kate	39089385	MM	1
175	Plitas, George	39102467	MC	1
198	Tadros, Audree	39091629	RW	1

complex_case_status	lesion_size_status_source	lesion_size_status_human
0	1	1
1	1	1
0	1	1
0	1	1
0	1	1
1	1	1
0	1	1
0	1	1
0	1	1
0	1	1
0	1	1
0	1	1
0	1	1
0	1	1
0	1	1

lesion_size_status_ai	laterality_status_source	laterality_status_human	laterality_status_ai
1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1
3	1	1	1
3	1	1	1
3	1	1	1
3	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1
3	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1

lesion_location_status_source	lesion_location_status_human	lesion_location_status_ai
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1

calcifications_asymmetry_status_source	calcifications_asymmetry_status_human
1	1
0	NA
1	1
1	1
0	NA
1	1
1	1
1	1
1	1
0	NA
1	1
0	NA
0	NA
0	NA
1	1

calcifications_asymmetry_status_ai	additional_enhancement_mri_status_source
1	1
NA	0
1	1
1	0
NA	0
1	1
1	0
1	1
1	0
NA	0
1	0
NA	0
NA	0
NA	0
1	0

additional_enhancement_mri_status_human	additional_enhancement_mri_status_ai
1	1
NA	NA
1	1
NA	NA
NA	NA
1	2
na	na
1	1
na	na
NA	NA
NA	NA
NA	NA
NA	NA
NA	NA
NA	NA

extent_status_source	extent_status_human	extent_status_ai
0	NA	NA
1	2	1
1	1	1
1	1	3
0	NA	NA
1	1	1
0	na	na
1	1	1
0	na	na
0	NA	NA
1	1	2
0	NA	NA
0	NA	NA
0	NA	NA
0	NA	NA

accurate_clip_placement_status_source	accurate_clip_placement_status_human
1	1
1	2
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
0	NA
1	1
1	1
1	1
1	1

accurate_clip_placement_status_ai	workup_recommendation_status_source
3	1
1	0
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	0
NA	0
1	0
1	1
1	1
1	1

workup_recommendation_status_human	workup_recommendation_status_ai
1	1
NA	NA
1	3
1	1
2	1
1	3
1	1
1	1
1	1
NA	NA
NA	NA
NA	NA
1	1
1	1
1	1

Lymph node_status_source	Lymph node_status_human	Lymph node_status_ai
1	1	1
0	NA	NA
1	1	1
1	1	1
1	1	1
1	1	1
0	na	na
1	1	2
1	2	1
1	1	1
1	1	1
1	1	1
1	1	1
0	NA	NA
1	1	1

chronology_preserved_status_source	chronology_preserved_status_human
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1

chronology_preserved_status_ai	biopsy_method_status_source	biopsy_method_status_human
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
3	1	1
1	1	1
1	1	1

biopsy_method_status_ai	invasive_component_size_pathology_status_source
1	0
1	1
1	0
1	0
1	1
1	0
1	1
1	1
1	1
1	1
1	1
1	0
1	1
3	1
1	1

invasive_component_size_pathology_status_human	invasive_component_size_pathology_status_ai
NA	NA
1	1
NA	NA
NA	NA
2	1
NA	NA
2	1
1	1
2	3
2	2
2	3
NA	NA
2	1
2	1
1	1

histologic_diagnosis_status_source	histologic_diagnosis_status_human
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1

histologic_diagnosis_status_ai	receptor_status_source	receptor_status_human
1	1	1
3	1	2
1	1	1
1	0	NA
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1

receptor_status_ai	v2 corrected
1	y
1	y
1	y
NA	y
1	n (0.7 instead of 0.6)
1	
2	
1	
1	
3	
1	
1	
1	
1	
3	

comments
clip not deployed/seen on post-biopsy images however AI reports "Yes (confirmed on subsequent imaging)". All imaging repo
Ai read it as lobular reports say mammary
AI correctly wrote no Her2 in the receptors section, but then called it a triple negative cancer is work up section - just DCIS
in describing extent of disease from diag mammo - AI calls area cals when report says densities
Size of tumor os US was 0.4, AI wrote 1.4, uploaded text not clear, may be reason for discepency
MRI was available but AI did not mention it. AIRecommended staging workup.
missed receptors and reported incorrect size of asymetry on MSK reivew (4.3 rather than 1.3)---clinically sigificant error
Bot gave incorrect size for US mass 1.5 instead of 0.5
Bot reports invasive component as 3.5mm, was 5.5mm
bot reports HR as TNBC, but triple positive. Bot also misses invasive component size. Source document long and blurry
Poor quality PDF in which bot missed extent of disease in US report and interpreted 1.1 as 1.7 in path report
Bot incorrectly reads initial left breast ultrasound as 6.5cm, when its 6mm. It does however identify this as a size discrepancy a
both human and AI HPI fail to mention no axillary lymphadenopathy. Perhaps because MSK review states that the axilla was n
bot got the receptors, stating ER- PR - when in fact they were ER+ PR+

AI Failure Analysis — what predicts AI=3 (severe error)?

15 of 200 rows (7.5%) had at least one field where AI got a severe error (status=3). This sheet analyzes what features distinguish those 15 rows from the other 185.

1. WHICH FIELDS THE AI FAILS ON

Field	# severe errors (AI=3)	% of failure events
lesion_size	5	31%
workup_recommendation	2	13%
invasive_component_size_pathology	2	13%
receptor	2	13%
extent	1	6%
accurate_clip_placement	1	6%
chronology_preserved	1	6%
biopsy_method	1	6%
histologic_diagnosis	1	6%
laterality	0	0%
lesion_location	0	0%
calcifications_asymmetry	0	0%
additional_enhancement_mri	0	0%
Lymph node	0	0%
TOTAL events (15 rows; row 71 had 2 failed field	**16**	**100%**

2. SURGEON — does failure cluster on specific surgeons?

Surgeon	n cases	n failures

El-Tamer, Mahmoud	10	4
Montag, Giacomo	10	2
Barrio, Andrea	10	1
Capko, Deborah	10	1
Heerdt, Alexandra	10	1
Kirstein, Laurie	10	1
Lee, Min	10	1
Moo, Tracy-Ann	10	1
Pawloski, Kate	10	1
Plitas, George	10	1
Tadros, Audree	10	1
(9 other surgeons)	90	0
TOTAL	**200**	**15**

3. CASE CHARACTERISTICS

Feature	Group	n cases
tumor_invasive_dcis	1 (Invasive)	164
tumor_invasive_dcis	2 (DCIS)	36
complex_case_status	0 (not complex)	162
complex_case_status	1 (complex)	38

4. SOURCE-DATA AVAILABILITY (does the AI fail when source is missing fields?)

Metric	Fail rows (n=15)	Pass rows (n=185)
Avg # fields with source=0 (not in source data)	2.67	2.49
Avg # fields with human=2 (human noted minor is	0.80	1.36
Avg # fields with AI=2 (AI minor issue)	0.33	0.39

5. HUMAN-vs-AI DISAGREEMENT — does the human catch the same things?

Of the 16 specific field-level severe errors flagged by AI (across 15 rows):

Human rated same field as 1 (no issue) — i.e. AI ur	14	88%
Human ALSO flagged same field (human=2 or 3)	2	13%

6. THE 16 DISAGREEMENT EVENTS (drill-down)

Row	Field	Source
13	accurate_clip_placem	1
32	histologic_diagnosis	1
63	workup_recommenda	1
64	extent	1
67	lesion_size	1
71	lesion_size	1
71	workup_recommenda	1
95	lesion_size	1
104	lesion_size	1
118	invasive_component_	1
135	receptor	1
138	invasive_component_	1
145	lesion_size	1
164	chronology_preserved	1
176	biopsy_method	1
199	receptor	1

BOTTOM LINE

1. The AI's failure mode is specific, not random: 9 of 16 severe errors (56%) are on numeric measurement fields (lesion_size, invasive_component_size_pathology) or clinical-judgment fields (workup_recommendation, receptor). It does not fail on simple categoricals (laterality, location, lymph node).

2. Failures cluster on two surgeons (El-Tamer 40%, Montag 20%) — accounting for 40% of all failures. Worth investigating: are these surgeons' note-writing styles harder to parse, or is the sample just small?

3. Source-data completeness, case complexity, and tumor type are NOT meaningful predictors of AI failure. The AI's mistakes are not driven by missing input data.

4. In 14 of 16 severe-error events, the human reviewer rated the same field as clean. These are the cases that most need a third look — to determine whether the AI is hallucinating errors or catching real ones the human missed.

Takeaway

lesion_size alone accounts for 31% of all severe AI errors. Numeric measurement fields (lesion_size, invasive_component_size_pathology) and clinical-judgment fields (workup_recommendation, receptor) dominate. The AI does NOT fail on simple categorical fields (laterality, location, lymph node).

Failure rate

Takeaway

40%	*El-Tamer fails at 40% — 4x the cohort baseline of 7.5%. Montag is 2x. With only 10 cases each the sample is small, but these two surgeons account for 6 of 15 failures (40%). Cases involving them deserve extra QC.*
20%	
10%	
10%	
10%	
10%	
10%	
10%	
10%	
10%	
10%	
0%	
8%	

n failures	Failure rate
11	6.7%
4	11.1%
13	8.0%
2	5.3%

Takeaway

Source-data completeness is NOT the issue — fail and pass rows have nearly identical numbers of missing source fields. The AI is not failing because data was unavailable; it had access and got it wrong.

Takeaway

In 14 of 16 severe-error events, the human reviewer rated the same field as clean (status=1). Either the AI is hallucinating errors that don't exist (false positives), or the AI is catching real errors the human missed. Either way, these 16 disagreements are the highest-value review targets in the entire dataset.

Human	AI
1	3
1	3
1	3
1	3
1	3
1	3
1	3
1	3
1	3
2	3
1	3
2	3
1	3
1	3
1	3
1	3

Takeaway

DCIS cases fail slightly more often (11% vs 7%), but the sample is small. Surprisingly, 'complex case' flag does NOT predict AI failure — failure rate is actually lower for complex cases. Complexity, as labeled, is not what trips the AI up.

Green-shaded rows: human ALSO flagged the field (human=2). Other 14 events: human said clean, AI said severe error — these are the highest-value disagreement cases to triage.

Column1	surgeon	mrn	patient_initials	tumor_invasive_dcis
162	Pawloski, Kate	39090606	NF	1
164	Pawloski, Kate	39086335	SA	1
168	Pawloski, Kate	39091900	NB	1
181	Sacchini, Virgilio	39097273	ST	1
197	Tadros, Audree	39011543	MS	1
200	Tadros, Audree	38310241	CL	2

complex_case_status	lesion_size_status_source	lesion_size_status_human
0	1	1
0	1	3
0	1	1
0	1	2
1	1	3
0	1	1

lesion_size_status_ai	laterality_status_source	laterality_status_human	laterality_status_ai
1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1

lesion_location_status_source	lesion_location_status_human	lesion_location_status_ai
1	1	1
1	1	1
1	1	1
1	2	1
1	1	1
1	1	1

calcifications_asymmetry_status_source	calcifications_asymmetry_status_human
1	1
1	2
1	1
1	2
0	NA
1	1

calcifications_asymmetry_status_ai	additional_enhancement_mri_status_source
1	0
1	0
2	1
1	0
NA	1
1	0

additional_enhancement_mri_status_human	additional_enhancement_mri_status_ai
NA	NA
NA	NA
1	1
NA	NA
1	2
NA	NA

extent_status_source	extent_status_human	extent_status_ai
0	NA	NA
0	NA	NA
1	3	1
0	NA	NA
1	1	2
0	NA	NA

accurate_clip_placement_status_source	accurate_clip_placement_status_human
1	1
1	1
1	1
1	2
1	1
1	2

accurate_clip_placement_status_ai	workup_recommendation_status_source
1	1
1	1
1	1
1	1
1	1
1	1

workup_recommendation_status_human	workup_recommendation_status_ai
2	2
2	1
2	2
2	2
1	1
1	1

Lymph node_status_source	Lymph node_status_human	Lymph node_status_ai
1	1	1
0	NA	NA
1	1	1
1	1	1
0	NA	NA
1	2	1

chronology_preserved_status_source	chronology_preserved_status_human
1	3
1	1
1	1
1	2
1	1
1	1

chronology_preserved_status_ai	biopsy_method_status_source	biopsy_method_status_human
1	1	1
1	1	1
1	1	1
1	1	2
1	1	1
1	1	3

biopsy_method_status_ai	invasive_component_size_pathology_status_source
1	0
1	0
1	1
1	1
1	0
1	0

invasive_component_size_pathology_status_human	invasive_component_size_pathology_status_ai
NA	NA
NA	NA
2	1
2	1
NA	NA
NA	NA

histologic_diagnosis_status_source	histologic_diagnosis_status_human
1	1
1	1
1	1
1	2
1	1
1	1

histologic_diagnosis_status_ai	receptor_status_source	receptor_status_human
1	0	NA
1	1	1
1	1	1
1	1	3
1	1	1
1	1	1

receptor_status_ai
NA
1
2
2
2
1